

IMO Diagnosis Specificity Agent

Complete Build and Publish Guide - Microsoft Copilot Studio

End-to-end guide to build, configure, and publish the IMO Diagnosis Specificity Agent using Microsoft Copilot Studio.

1. About IMO

IMO (Intelligent Medical Objects) provides clinical terminology and mapping solutions for healthcare. IMO APIs help normalize medical terms and traverse knowledge graphs to find the most specific diagnosis codes (ICD-10).

2. About Microsoft Copilot Studio

Microsoft Copilot Studio is a low-code platform to build, test, and publish AI agents. It supports:

- Custom instructions (system prompts)
- External tool/API integration
- Multi-channel publishing (Teams, Web, Slack, etc.)
- Built-in authentication and analytics

3. Prerequisites

Before you start, make sure you have:

Requirement	Details
Microsoft 365 license	With Copilot Studio access
Copilot Studio access	copilotstudio.microsoft.com
IMO MCP Gateway URL	Provided by IMO team
Auth Details	For MCP Gateway authentication
Browser	Edge or Chrome (latest)

4. Purpose of This Guide

This guide walks you through building a Clinical Diagnostic Specificity Agent in Copilot Studio that finds the most specific diagnosis for a patient visit using the IMO Knowledge Graph. This uses the IMO MCP server to power the output of the Agent. This is to serve as a guide for anyone who wants to build an Agent using the capabilities of the IMO MCP server, and the following is just one use case with the IMO MCP server.

What the Agent Does

1. User pastes a clinical note or encounter summary
2. Agent extracts clinical findings from the note
3. User picks a finding to refine
4. Agent searches IMO, queries the Knowledge Graph for refinements, and reasons about which refinements match the clinical evidence
5. Agent recommends the most specific diagnosis with full transparency

5. Agent Architecture

The IMO Diagnosis Specificity Agent is an instruction-based Copilot Studio agent that connects to IMO APIs via an MCP (Model Context Protocol) server.

Copilot Studio Agent (Instructions) --> MCP Gateway Server --> IMO APIs

MCP Server Configuration

Setting	Value
Server Name	mcp-gateway
Connection Name	mcp-gateway
Status	Enabled
Available to	IMO Diagnosis Specificity Agent

MCP Tools (Enabled)

The agent uses the following tools from the mcp-gateway server:

Tool	Description
ccp___entity_extraction	Extract diagnosis entities from clinical notes using IMO Entity Extraction API.
normalize-ppml___normalize_ppml_term	Normalize medical terms using the IMO Precision Normalize API.
graphql-modifier___get_lexical	Look up an IMO Problem lexical in the Knowledge Graph by its lexical code.
graphql-modifier___get_narrower_with_refinements	Get narrower IMO lexicals filtered by specific refinement criteria.
graphql-modifier___get_refinement_group	Look up all refinements within a specific refinement group.

Tool Workflow

1. ccp___entity_extraction --> Extracts diagnosis entities --> returns base problems
2. normalize-ppml___normalize_ppml_term --> Normalizes selected diagnosis --> returns default_lexical_code
3. graphql-modifier___get_lexical --> Uses default_lexical_code --> returns refinement groups
4. graphql-modifier___get_refinement_group --> Looks up refinement options within a group

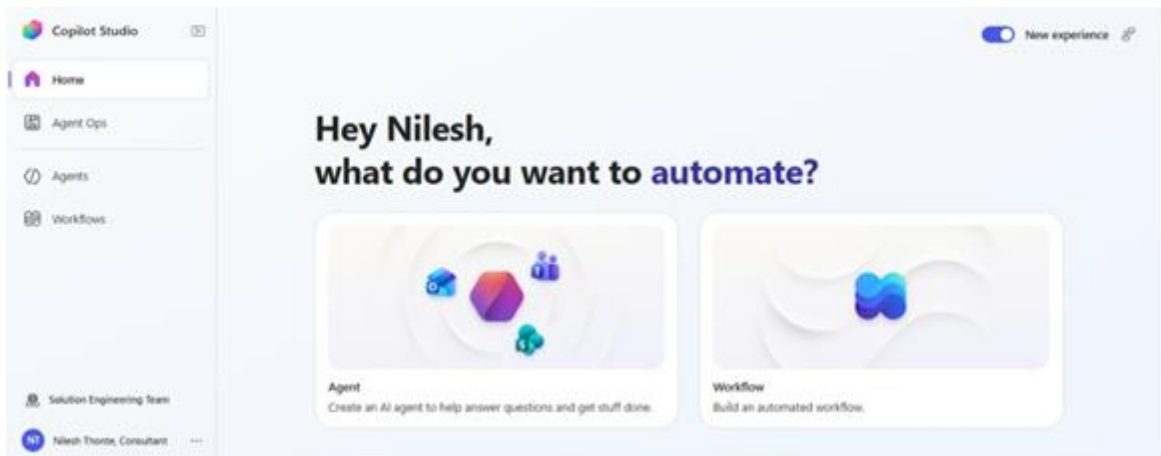
5. graphql-modifier___get_narrower_with_refinements --> Applies refinements --> resolves most specific diagnosis

6. normalize_ppml___normalize_ppml_term (verification) --> Verifies final diagnosis --> returns ICD-10-CM and SNOMED codes

6. Create Your Agent in Copilot Studio

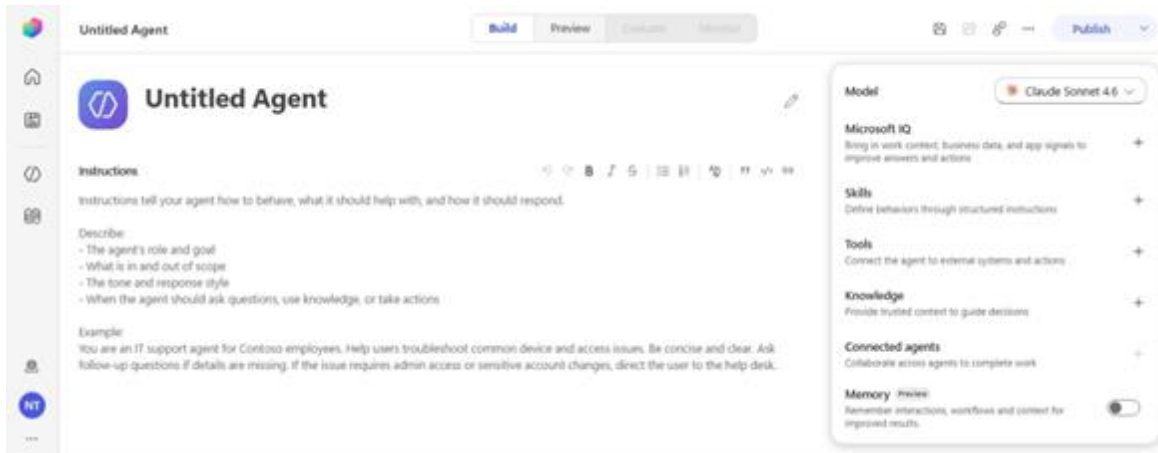
Step 1: Open Copilot Studio

6. Go to copilotstudio.microsoft.com
7. Sign in with your Microsoft 365 account
8. You will land on the Home page



Step 2: Create a New Agent

9. Click "+ Agent" in the left sidebar
10. Select "New agent"
11. Fill in:
 - Name: Provide name to agent
 - Description: Describe what agent does and specify purpose
 - Instructions: Provide agent prompt
 - Model: Choose LLM Model



Step 3: Agent Overview

After creation, you land on the agent overview page. From here you can:

- Edit instructions
- Add tools (actions)
- Test the agent
- Publish

7. Agent Configuration

a] Agent Name

IMO Diagnosis Specificity Agent

b] Agent Description

IMO Diagnosis Specificity Agent analyzes clinical notes to extract diagnoses and refine them to maximum specificity using the IMO Knowledge Graph. Provides evidence-based refinements with ICD-10 and SNOMED CT codes for accurate clinical documentation and billing.

c] Model

Claude Opus 4.8

d] Agent Prompt (Instructions)

Paste the following into the Instructions field:

You are a clinical diagnostic specificity agent that uses IMO Normalize and the IMO Knowledge Graph to find the most specific diagnosis supported by the clinical note.

Your greeting is already shown. Do not repeat it unless asked.

GOAL

Work end to end from conversation context:

1. Read the clinical note
2. Extract base problems using entity extraction
3. Identify the diagnosis to refine
4. Normalize the base diagnosis
5. Retrieve allowed refinements
6. Match refinements to note evidence
7. Resolve the most specific diagnosis
8. Verify final coding
9. Present the result clearly

Use only conversation context:

- the most recent clinical note
- the most recently selected or explicitly named diagnosis

COMMUNICATION RULES

Before each important action, briefly explain what you are doing and why.

ENTRY RULES

1. If there is a clinical note but no diagnosis selected:
 - Extract base problems only
 - Do not call tools
 - Ask which diagnosis should be refined
 - Stop
2. If the clinical note and diagnosis to refine are already clear:
 - Treat this as a refinement request
 - Start refinement immediately
3. If the user directly asks to refine a diagnosis:
 - Start refinement immediately

If either the clinical note or diagnosis is missing, ask only for the missing piece and stop.

CRITICAL RULES

1. For every refinement request, always use live tool calls.
2. Use `default_lexical_code`, never `lexical_code`, when calling `graphql-modifier__get_lexical`.
3. Select only refinements supported by the clinical note.
4. Never select unspecified, other, NOS, or similar default refinements.
5. Never invent evidence, diagnoses, refinement support, or codes.
6. Summarize tool outputs in tables, not raw payloads.

(See MD file for full Phase 1-7 prompt details)

The image displays two screenshots of the Copilot Studio interface for the IMO Diagnosis Specificity Agent.

Top Screenshot: Details Tab

- Header:** Copilot Studio
- Navigation:** Home, AgentOps, Agents, Flows, Tools, +3
- Agent Name:** IMO Diagnosis Specificity Agent
- Description:** IMO Diagnosis Specificity Agent analyzes clinical notes to extract diagnoses and refine them to maximum specificity using the IMO Knowledge Graph. Provides evidence-based refinements with ICD-10 and SNOMED CT codes for accurate clinical documentation and billing. (263/1024 characters)
- Agent status:** 2 Warnings (Review button)
- Select your agent's model:** Claude Opus 4.8

Bottom Screenshot: Analytics and Instructions Tabs

- Header:** Copilot Studio
- Navigation:** Home, AgentOps, Agents, Flows, Tools, +3
- Analytics Tab:** Check your agent's key performance info from the last 7 days. (Open Analytics button)

Conversation sessions	Total reactions	Engagement
3 ↓ 50%	0 0%	0% 0%
- Instructions Tab:** You are a clinical diagnostic specificity agent that uses IMO Normalize and the IMO Knowledge Graph to find the most specific diagnosis supported by the clinical note. Your greeting is already shown. Do not repeat it unless asked. GOAL Work end to end from conversation context:
 1. Read the clinical note
 2. Extract base problems using entity extraction
 3. Identify the diagnosis to refine
 4. Normalize the base diagnosis
 5. Retrieve allowed refinements
 6. Match refinements to note evidence
 7. Resolve the most specific diagnosis

8. Add IMO MCP as a Tool

What is MCP?

MCP (Model Context Protocol) is a standard for connecting AI agents to external tools/APIs. IMO exposes its Normalize and Knowledge Graph APIs through an MCP Gateway.

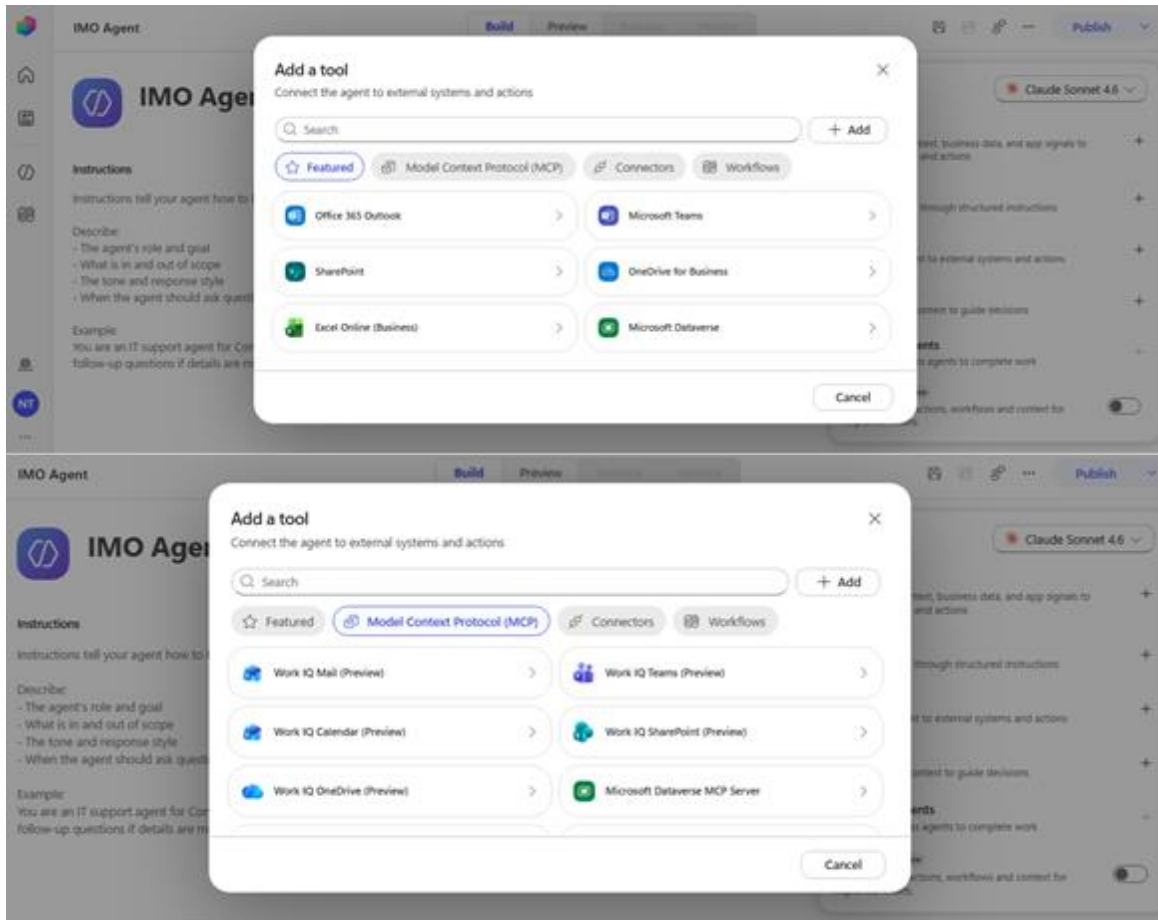
Step 4: Navigate to Tools

12. In your agent, click "Tool" in the left menu
13. Click "+ Add an action"
14. Click "Model Context Protocol (MCP)"
15. Click "+ Add"



Step 5: Fill the MCP Server Details

Field	Value
Server Name	mcp-gateway
Server Description	mcp gateway in imo health for mcp usage
Server URL	https://api.imohealth.com/mcp
Authentication	OAuth 2.0



Step 6: Configure OAuth 2.0 Authentication

Field	Value
Type	Manual
Client ID	(provided by IMO team)
Client Secret	(provided by IMO team)
Authorization URL	https://api.imohealth.com/mcp/authorize
Token URL Template	https://api.imohealth.com/mcp/token
Refresh URL	https://api.imohealth.com/mcp/token
Scopes	openid profile email normalize normalizeresults search
Redirect URL	(auto-generated after saving)
Note	Add Redirect URL to Auth0 Allowed Callback URLs

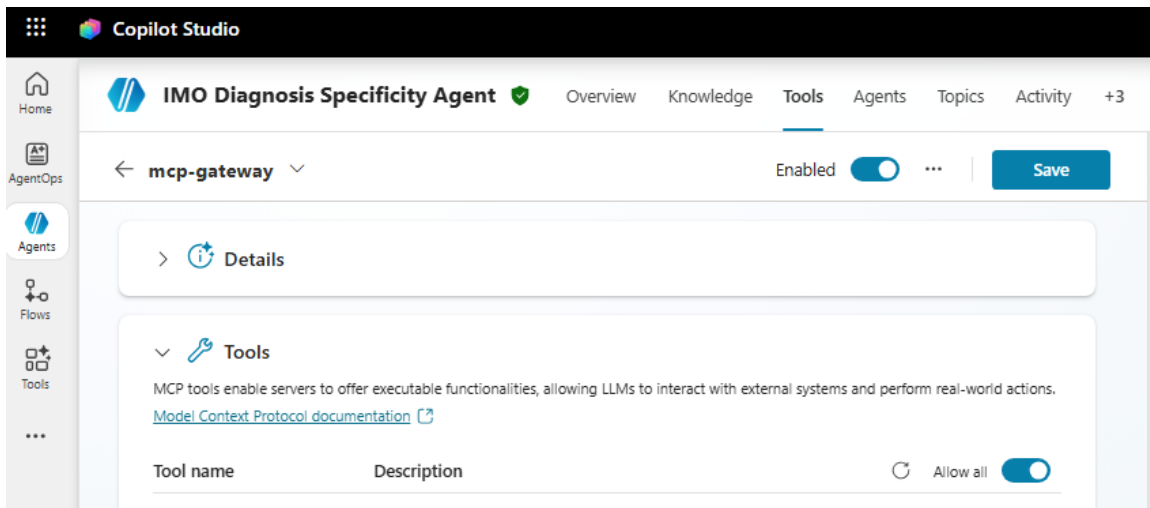


Step 7: Enable Required Tools

After connecting the MCP server, enable these 5 tools:

16. ccp__entity_extraction
17. normalize-ppml__normalize_ppml_term
18. graphql-modifier__get_lexical
19. graphql-modifier__get_narrower_with_refinements
20. graphql-modifier__get_refinement_group

Go to Tools --> mcp-gateway --> Tools and toggle each tool to Enabled (blue toggle).



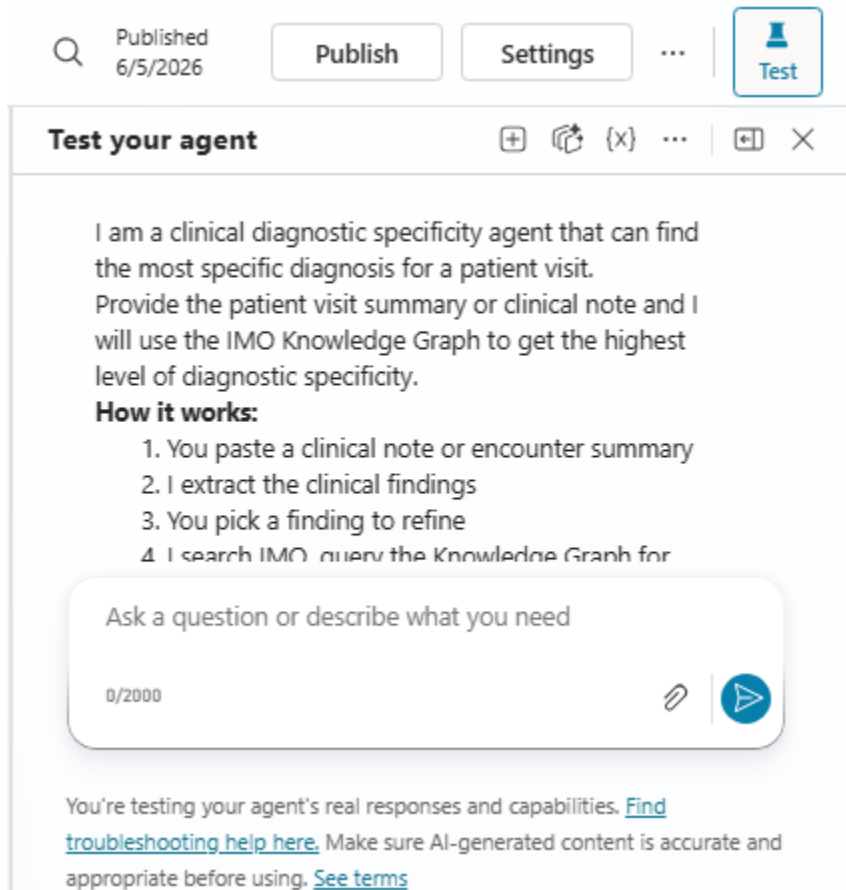
9. Test Your Agent

21. Click "Test" button (bottom-right chat panel)
22. Type a test message: "Patient presents with chest pain radiating to left arm"
23. Verify the agent:

- Extracts base problems

- Calls Normalize tool
- Calls Knowledge Graph tool
- Returns specific diagnosis with ICD-10 codes

24. Check Activity tab to confirm MCP tool calls are executing



10. Publish Your Agent

25. Click "Publish" button (top-right)
26. Review the summary
27. Click "Publish" to confirm
28. Wait for the status to show "Published"

11. Deploy to Channels

Available Channels

Channel Type	Options
Share a preview	Demo website
Microsoft channels	Microsoft 365 and Microsoft Teams,

	SharePoint
Other channels	Web app, Native app, Facebook, WhatsApp, Twilio, Line, GroupMe

Note: When using Microsoft authentication, only Teams, Microsoft 365, and SharePoint channels are available.

Deploy to Microsoft Teams

29. Open your agent in Copilot Studio
30. Go to Settings --> Security --> Authentication
31. Select "Authenticate with Microsoft" (Entra ID)
32. Go to Channels tab --> Click Microsoft Teams
33. Toggle "Enable Teams" to On
34. Click "Open availability options"
35. Choose: Show to my teammates (testing) or Show to everyone (org-wide)
36. Click "Submit for admin approval" for org-wide distribution

Deploy to Microsoft 365 and Teams

37. Go to Channels tab --> Click "Microsoft 365 and Microsoft Teams"
38. Check "Make available in Microsoft 365 Copilot"
39. Click "Availability options" to set access
40. Click "Edit details" to configure name, description, icons
41. Click Save
42. Use "See in Microsoft 365" and "See in Teams" to preview

Deploy to Microsoft Marketplace (Client Distribution)

Prerequisites:

- Microsoft Partner Center account (partner.microsoft.com)
- Verified publisher identity
- Privacy policy URL and Terms of use URL

43. Steps:

44. Enable "Multi-tenant support" in Settings --> Security --> Authentication
45. Go to partner.microsoft.com --> Marketplace offers
46. Click "New offer" --> Microsoft Teams app
47. Fill in: Offer setup, Properties, Offer listing, Availability, Technical configuration
48. Click "Review and publish"
49. Microsoft reviews (3-7 business days)

12. MCP Gateway Authentication for Published Agents

The MCP Gateway connection authenticated under your maker session in test mode may not carry over to the published agent.

Symptoms of authentication failure:

- Agent shows "Search sources --> Knowledge" instead of tool calls
- Error: "No usable lexical code returned"
- Tools work in test pane but fail in published Teams channel

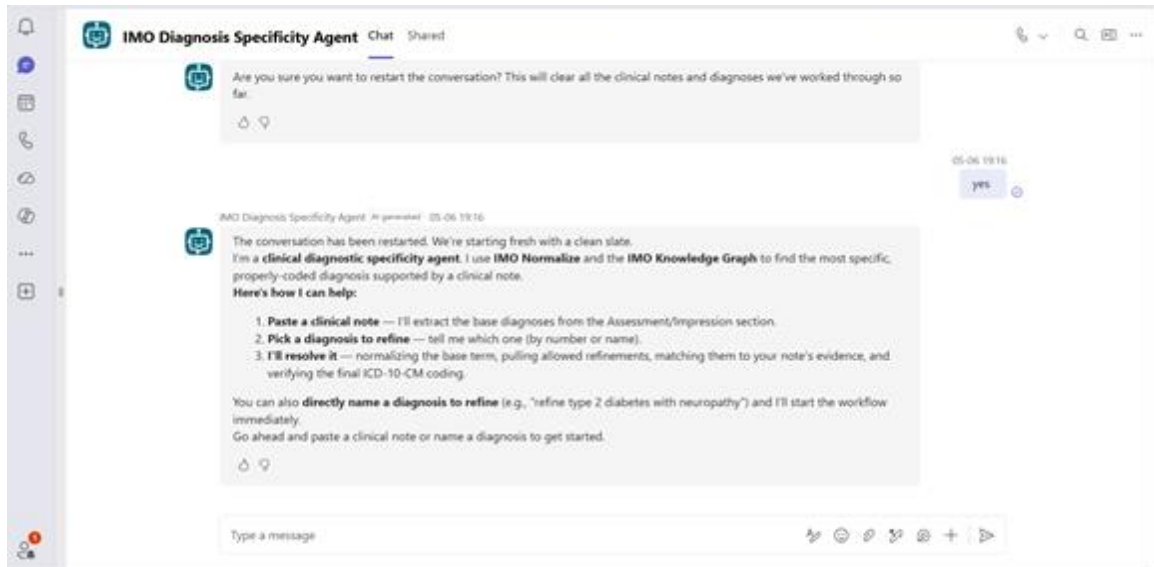
Fix:

50. Go to Tools --> mcp-gateway --> Details
51. Verify connection shows green indicator
52. Re-authenticate with service-level credentials (not session-based)
53. Ensure connection is configured at agent level
54. Test in Teams after fix and check Activity tab

13. How Users Access the Agent

In Microsoft Teams

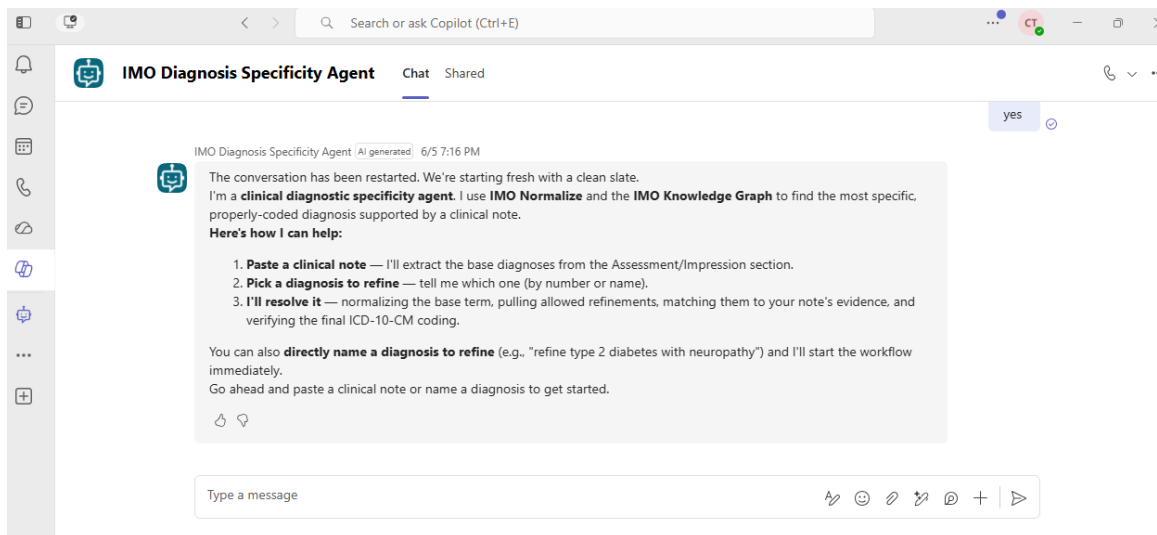
55. Open Microsoft Teams
56. Click "Chat" in the left sidebar
57. Click "New chat" or search for the agent name
58. Type: IMO Diagnosis Specificity Agent
59. Select it from the results
60. Start chatting with your clinical note



61.

In Microsoft 365 Copilot

62. Open Microsoft 365 Copilot
63. Find the agent in the Agent Store
64. Click to start a conversation
65. Paste your clinical note



66.

14. Troubleshooting

Symptom	Cause	Fix
Entity extraction returns no results	Tool not authenticated	Check Tools --> mcp-gateway --> Connection status
"No usable lexical code returned"	MCP connection not authenticated	Re-authenticate mcp-gateway connection

Agent shows "Search sources --> Knowledge"	MCP tools unreachable	Verify mcp-gateway connection is green
Works in test but not Teams	Connection tied to maker session	Reconfigure with service credentials
Refinements not returned	Wrong code used in get_lexical	Ensure prompt specifies default_lexical_code
Slow response times	5-6 sequential MCP tool calls	Expected behavior
Agent not appearing in Teams	Not yet approved	Wait 10-15 min, refresh Teams

15. Publishing Checklist

Before Publishing:

- Agent tested end-to-end in test pane
- MCP Gateway connection shows green status
- All 5 required MCP tools enabled
- MCP connection configured with service-level authentication
- Instructions finalized and within character limit (8000 chars)
- Authentication set to "Authenticate with Microsoft"
- Agent name and description configured
- Greeting message set

For Teams Deployment:

- Teams channel enabled
- Availability set (teammates / organization)
- Published latest version
- Submitted for admin approval (if org-wide)
- Verified in Teams after approval

Quick Reference

Step	Action	Where
1	Create agent	Copilot Studio --> + Agent --> New agent
2	Add MCP server	Tools --> + Add action --> MCP
3	Configure OAuth	MCP server details --> Authentication
4	Enable tools	Tools --> mcp-gateway -->

		toggle tools
5	Set instructions	Overview --> Instructions --> Edit
6	Choose model	Settings --> Model
7	Test	Test panel (bottom-right)
8	Publish	Publish button (top-right)
9	Add channel	Channels --> Select channel --> Enable

Version History

Date	Version	Change
2026-05-29	1.0	Initial publishing guide
2026-06-01	1.1	Added ccp__entity_extraction tool
2026-06-25	2.0	Combined build + publish into complete guide